BALRAJ PARSI

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LinkedIn | Portfolio | GitHub

PROFESSIONAL SUMMARY

Backend Software Engineer with 6+ years at Amazon, Microsoft, and Capital One. Build and operate Java microservices at scale, from Kafka-based data pipelines processing 100K+ daily changes to deployment infrastructure supporting millions of hosts on AWS and Azure.

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript
Frameworks & Libraries: Spring Boot, JUnit, TestNG, Mockito, Node.js
Databases: DynamoDB, Aurora DB (PostgreSQL), MySQL
Messaging & Streaming: Apache Kafka, Kafka Streams, AWS SQS, AWS SNS
AWS: EC2, S3, Lambda, CloudFormation, Route 53, Secrets Manager, CloudWatch, CodeDeploy, VPC
CI/CD & Tools: Jenkins, JFrog Artifactory, SonarQube, Mend (WhiteSource), Docker, Git, PagerDuty
Practices: Microservices Architecture, RESTful APIs, OAuth2, Agile/Scrum, CI/CD, On-Call Production Support

PROFESSIONAL EXPERIENCE

Backend Software Engineer | Capital One Jun 2025 - Present
Richmond, VA

- Own and maintain 4 Java microservices across 15 repositories for the Small Business Bank (SBB) platform. Handle development, testing, and production deployments with a 3-person engineering team and no dedicated QA.
- Built Kafka Streams data pipelines that sync customer PII between Systematics and RA through nightly batch jobs, processing 80K-100K record changes daily and removing data duplication at the enterprise level.
- Designed and implemented a bidirectional sync between Business Core and Systematics, which cut duplicate records and reduced API hit times for downstream services.
- Led the database migration from Aurora DB to DynamoDB, restructuring data access patterns to support the SBB modernization initiative.
- Resolved multiple production defects across services, including a VP-escalation incident where address updates were nullifying customer name fields. Deployed a hotfix within 40 minutes, limiting impact to 9 customers out of ~100K daily transactions.
- Replaced .jks certificate-based authentication with Turing OAuth2 configuration across service repositories, removing the need for manual certificate rotation every 90 days.
- Added cyber audit logging to DynamoDB tables to satisfy Capital One's security and compliance requirements for customer data access tracking.
- Run CI/CD pipelines through Jenkins and One Pipeline with JFrog Artifactory for artifact management, SonarQube for code quality gates, and Mend for dependency vulnerability scanning. Average 2-3 production deployments per week using AWS CodeDeploy.

Software Engineer (Contract) | Microsoft Jan 2024 - May 2025
Redmond, WA

- Owned backend services (Notification Service, PartialSync, Multiplexer) for Azure Resource Graph (ARG), handling data synchronization and query processing across 5+ global regions.
- Optimized real-time data sync logic across ARG backend services, reducing recurring synchronization failures and improving query performance for large-scale resource lookups.
- Tuned Global Traffic Manager (GTM) routing for ARG, rebalancing cross-region traffic distribution across WEU and SEA regions to reduce outage frequency and improve failover reliability.
- Wrote PowerShell and C# validation scripts to catch configuration drift and data inconsistencies before production rollouts, reducing deployment failures by 20%.
- Migrated CI/CD pipelines from Classic to YAML-based Azure DevOps, cutting review cycle times by 40%. Recognized by senior leadership for the efficiency improvement.
- Built KQL-based monitoring dashboards and alerting for backend service health, using log analysis to diagnose production issues and track data synchronization failures in real time.

- Authored a Technical Support Guide for non-Regional Agnostic scale units that reduced onboarding time for new engineers by 35%.

Software Development Engineer | Amazon Web Services Jun 2022 - Dec 2023

Seattle, WA

- Built and maintained features for Apollo, Amazon's internal deployment platform that ships software to Tier-1 services (S3, DynamoDB, Amazon Retail) across 6 million+ managed EC2 hosts.
- Designed and deployed RESTful APIs using Java, Spring Boot, and Jackson, with full test coverage through JUnit and Mockito. Owned the lifecycle from design through production release.
- Created load-balancing microservices in Node.js for partial fleet deployments, striping requests across load balancers and availability zones to maximize uptime.
- Migrated on-premise applications to AWS (S3, EC2, RDS, SQS, SNS, CloudFormation, VPC) and added CloudWatch and New Relic monitoring, reducing system downtime by 30%.
- Automated cleanup of stale container images from Amazon Container Repository using Coral-based integration tests, preventing build failures caused by outdated artifacts.
- Built and enhanced Apollo Import (Transport pipeline) by writing comprehensive unit tests for the Apollo Client, resolving failing test cases and improving deployment reliability.
- Handled on-call production support rotations, triaging and resolving critical system issues to minimize customer impact across high-traffic Amazon services.

Sr. Software Engineer | HCL Technologies Ltd. Jul 2019 - Aug 2021

Chennai, India

- Developed Python data processing modules for Cisco's CA-BOLT project, optimizing 350+ generic modules and automating code quality checks with custom scripts.
- Built real-time alerting systems in Python that notified customers when product issues were detected, cutting incident response times across Cisco's support workflow.
- Maintained and extended legacy PHP backend applications, fixing security vulnerabilities aligned with OWASP standards and implementing input validation to prevent injection attacks.
- Configured Jenkins CI/CD pipelines with Groovy scripts and integrated automated test execution for Java and Node.js applications.
- Managed datasets from Cisco products (NXOS, iOS, ACI, VPN, AnyConnect) and handled data integration workflows to maintain operational accuracy.
- Developed internal monitoring tools using Python to automate log management and system health checks, improving operational efficiency by 30% across the team.

EDUCATION

Executive MS in Artificial Intelligence Oct 2025 - Present

University of the Cumberlands

MS in Computer Science, Data Science (GPA: 3.83) Aug 2021 - Dec 2022

University of Cincinnati

B.Tech in Electronics and Communication Engineering (GPA: 9.6) Aug 2015 - Apr 2019

Anurag Group of Institutions (formerly CVSR), Hyderabad